

Causal Inference

• Science wants to discover causes, not just patterns.

• But causes are tricky because: 1) we cannot directly see causation (Hume's point) 2) we only see events happening together.

• So scientists must infer causes from data.

There are two types of causal claims: 1) Cause of a single event → meteor caused dinosaur extinction 2) General causal law → smoking causes lung cancer.

Here we focus on the second type

Q) What is causal claim in science?

• When we say: Smoking causes lung cancer we are not saying: 1) every smoker must get cancer 2) cancer cannot occur without smoking 3) It is logically unavoidable.

Instead, we mean:

- 1) Smoking significantly increases the probability of lung cancer
- 2) It is a major contributing factor
- 3) there is a causal mechanism (cell damage, mutations, etc.)

This is empirical causation, not logical necessity

January 2024							February 2024						
Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa	Su
01	1	2	3	4	5	6	7	05			1	2	3
02	8	9	10	11	12	13	14	06	5	6	7	8	9
03	15	16	17	18	19	20	21	07	12	13	14	15	16
04	22	23	24	25	26	27	28	08	19	20	21	22	23
05	29	30	31					09	26	27	28	29	